

INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION

Chapter 7

Sampling Distributions

1

兩個主要的分配

1. $X \sim N(\mu, \sigma^2)$

2. $X \sim \text{Bernoulli}(p) = \text{Binomial}(n = 1, p)$

課本隨機變數(a random variable)和可能出現的值(a realized value)都是用小寫！

p.171

缺點：需由語意判斷

DEFINITION

Let x be a discrete random variable with probability distribution $p(x)$. The mean or **expected value of x** is given as

$$\mu = E(x) = \sum x p(x)$$

summing over all the values of x .

We could use a similar argument to justify the formulas for the **population variance σ^2** and the **population standard deviation σ** . These numerical measures describe the spread or variability of the random variable using the “average” or “expected value” of $(x - \mu)^2$, the squared deviations of the x -values from their mean μ .

DEFINITION

Let x be a discrete random variable with probability distribution $p(x)$ and mean μ . The **variance of x** is

$$\sigma^2 = E[(x - \mu)^2] = \sum (x - \mu)^2 p(x)$$

summing over all the values of x .[†]

DEFINITION

The **standard deviation σ** of a random variable x is equal to the positive square root of its variance.

RECALL

- Data (Ch 1-3)

- 母體(population): $x_1, x_2, \dots, x_N$

- $\mu = \frac{1}{N} \sum_{i=1}^N x_i$

- $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$

- 樣本(sample): $x_1, \dots, x_n$

- $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

- $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

- Probabilty (Ch 4 - 7)

- 母體(population): X

- $\mu = E[X] = \sum xP(X=x) = \sum xp(x)$

- $\sigma^2 = E[(X-\mu)^2] = \sum (x-\mu)^2 P(X=x) = \sum (x-\mu)^2 p(x)$

- 樣本(sample): $X_1, \dots, X_n \sim X$

- $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

- $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

INTRODUCTION

- Parameters are numerical descriptive measures for populations.
 - For the normal distribution, the location and shape are described by μ and σ .
 - For a binomial distribution consisting of n trials, the location and shape are determined by p .
- Often the values of parameters that specify the exact form of a distribution are unknown.
- You must rely on the **sample** to learn about these parameters.

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7.1 SAMPLING PLANS AND EXPERIMENTAL DESIGNS

6

SAMPLING

Examples:

- A pollster is sure that the responses to his “agree/disagree” question will follow a binomial distribution, but p , the proportion of those who “agree” in the population, is unknown.
- An agronomist believes that the yield per acre of a variety of wheat is approximately normally distributed, but the mean μ and the standard deviation σ of the yields are unknown.
- ✓ If you want the sample to provide reliable information about the population, you must select your sample in a certain way!

SIMPLE RANDOM SAMPLING

- The **sampling plan** or **experimental design** determines the amount of information you can extract, and often allows you to measure the **reliability of your inference**.
- **Simple random sampling** is a method of sampling that allows each possible sample of size n an equal probability of being selected.

EXAMPLE



• There are 89 students in a statistics class. The instructor wants to choose 5 students to form a project group. How should he proceed?

1. Give each student a number from 01 to 89.
2. Choose 5 pairs of random digits from the random number table.
3. If a number between 90 and 00 is chosen, choose another number.
4. The five students with those numbers form the group.

06907	11008	42751	27756	53498
72905	56420	69994	98872	31016
91977	05463	07972	18876	20922
14342	63661	10281	17453	18103
36857	53342	53988	53060	59533
69578	88231	33276	70997	79936
40961	48235	03427	49626	69445
93969	52636	92737	88974	33488
61129	87529	85689	48237	52267
97336	71048	08178	77233	13916

TYPES OF SAMPLES

- Sampling can occur in two types of practical situations:

1. **Observational studies:** The data existed before you decided to study it. Watch out for
 - ✓ **Nonresponse:** Are the responses biased because only opinionated people responded?
 - ✓ **Undercoverage:** Are certain segments of the population systematically excluded?
 - ✓ **Wording bias:** The question may be too complicated or poorly worded.

TYPES OF SAMPLES

- Sampling can occur in two types of practical situations:

2. **Experimentation:** The data are generated by imposing an experimental condition or treatment on the experimental units.
 - ✓ **Hypothetical populations** can make random sampling difficult if not impossible.
 - ✓ Samples must sometimes be chosen so that the experimenter believes they are **representative** of the whole population.
 - ✓ Samples must **behave like random samples!**

OTHER SAMPLING PLANS

- There are several other sampling plans that still involve **randomization**:

1. **Stratified random sample:** Divide the population into subpopulations or **strata** and select a simple random sample from each strata.
2. **Cluster sample:** Divide the population into subgroups called **clusters**; select a simple random sample of clusters and take a census of every element in the cluster.
3. **1-in-k systematic sample:** Randomly select one of the first k elements in an ordered population, and then select every k -th element thereafter.

EXAMPLES



- Divide California into counties and take a simple random sample within each county. **Stratified**
- Divide California into counties and take a simple random sample of 10 counties. **Cluster**
- Divide a city into city blocks, choose a simple random sample of 10 city blocks, and interview all who live there. **Cluster**
- Choose an entry at random from the phone book, and select every 50th number thereafter. **1-in-50 Systematic**

NON-RANDOM SAMPLING PLANS

- There are several other sampling plans that do not involve **randomization**. They should **NOT** be used for statistical inference!

1. **Convenience sample:** A sample that can be taken easily without random selection.
 - People walking by on the street
2. **Judgment sample:** The sampler decides who will and won't be included in the sample.
3. **Quota sample:** The makeup of the sample must reflect the makeup of the population on some selected characteristic.
 - Race, ethnic origin, gender, etc.

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7.2 STATISTICS AND SAMPLING DISTRIBUTIONS

15

SAMPLING DISTRIBUTIONS

- Numerical descriptive measures calculated from the sample are called **statistics: $f(X_1, \dots, X_n)$ i.e., a function of random samples**
- Statistics vary from sample to sample and hence are random variables.
- The probability distributions for statistics are called **sampling distributions**.
- In repeated sampling, they tell us what values of the statistics can occur and how often each value occurs.

EXAMPLES OF STATISTICS

7.2

Statistics and Sampling Distributions

When you select a random sample from a population, the numerical descriptive measures you calculate from the sample are called **statistics**. These statistics vary or change for each different random sample you select; that is, they are *random variables*. The probability distributions for statistics are called **sampling distributions** because, in repeated sampling, they tell us:

- What values of the statistic can occur.
- How often each value occurs.

- Sample mean: $\bar{X} = \frac{X_1 + \dots + X_n}{n}$
- Sample variance: $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- Range: $R = \max(X_1, \dots, X_n) - \min(X_1, \dots, X_n)$
- Median, Q1, Q3

SAMPLING DISTRIBUTIONS

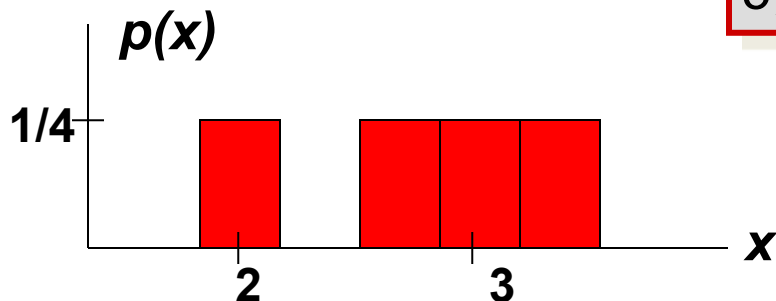
Definition: The **sampling distribution of a statistic** is the probability distribution for the possible values of the statistic that results when random samples of size n are repeatedly drawn from the population.

Population: 3, 5, 2, 1

Draw samples of size $n = 3$ *without replacement*

Possible samples	$\bar{x}$
3, 5, 2	$10/3 = 3.33$
3, 5, 1	$9/3 = 3$
3, 2, 1	$6/3 = 2$
5, 2, 1	$8/3 = 2.67$

Each value of $\bar{x}$ is equally likely, with probability $1/4$



$$\left\{ \begin{array}{l} P(\bar{X} = 3.33) = 0.25 \\ P(\bar{X} = 3) = 0.25 \\ P(\bar{X} = 2) = 0.25 \\ P(\bar{X} = 2.67) = 0.25 \end{array} \right.$$

IN-CLASS EXERCISE EXAMPLE 7.3

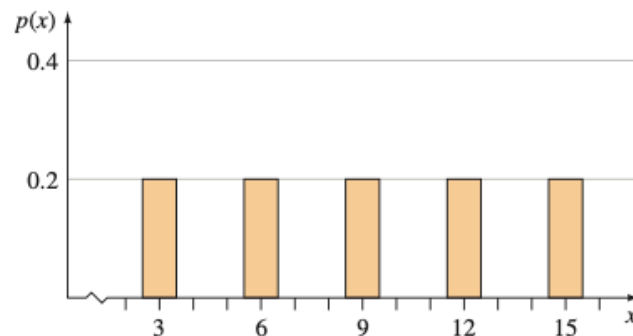
EXAMPLE 7.3

A population consists of $N = 5$ numbers: 3, 6, 9, 12, 15. If a random sample of size $n = 3$ is selected **without replacement**, find the sampling distributions for the sample mean $\bar{x}$ and the sample median m .

Solution We are sampling from the population shown in Figure 7.1. It contains five distinct numbers and each is equally likely, with probability $p(x) = 1/5$. We can easily find the population mean and median as

$$\mu = \frac{3 + 6 + 9 + 12 + 15}{5} = 9 \quad \text{and} \quad M = 9$$

Figure 7.1
Probability histogram for
the $N = 5$ population values
in Example 7.3



IN-CLASS EXERCISE

- Find the sampling distribution for the sample mean $\bar{X}$.
- Find the sampling distribution for the sample median M .
- Hint: Find the sample space of random samples with $n = 3$ without replacement.

■ **Table 7.3** Values of $\bar{x}$ and m for Simple Random Sampling when $n=3$ and $N=5$

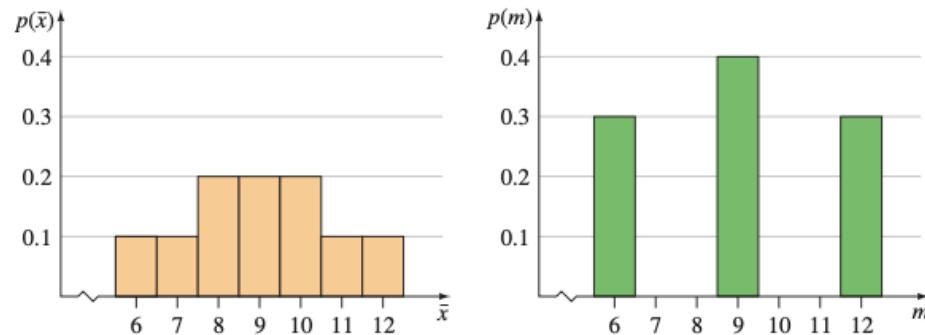
Sample	Sample Values	$\bar{x}$	m
1	3, 6, 9	6	6
2	3, 6, 12	7	6
3	3, 6, 15	8	6
4	3, 9, 12	8	9
5	3, 9, 15	9	9
6	3, 12, 15	10	12
7	6, 9, 12	9	9
8	6, 9, 15	10	9
9	6, 12, 15	11	12
10	9, 12, 15	12	12

■ **Table 7.4** Sampling Distributions for (a) the Sample Mean and (b) the Sample Median

(a)	$\bar{x}$	$p(\bar{x})$	(b)	m	$p(m)$
	6	.1		6	.3
	7	.1		9	.4
	8	.2		12	.3
	9	.2			
	10	.2			
	11	.1			
	12	.1			

Figure 7.2

Probability histograms for the sampling distributions of the sample mean, $\bar{x}$, and the sample median, m , in Example 7.3



IN-CLASS EXERCISE

- Population: $[-1, 0, 2]$ with probability $1/3$.
- Consider a sample of size $n = 2$. Draw a random sample **with replacement**.
- Find the sampling distribution of $\bar{X} = \frac{1}{2}(X_1 + X_2)$.
- Hints:
 - (a) List the sample space of (X_1, X_2) .
 - (b) Summarize the distribution of $\bar{X}$.

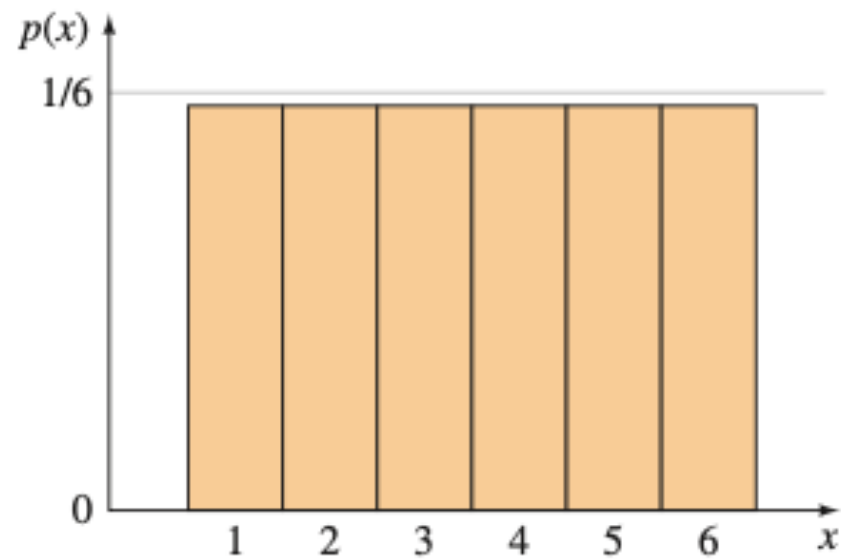
The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, with smaller circles below it, some of which are partially overlapping the stripes.

7.3 THE CENTRAL LIMIT THEOREM AND THE SAMPLE MEAN

INTUITION

Figure 7.3

Probability distribution for x , the number appearing on a single toss of a die



■ **Table 7.5(a) Sums of the Upper Faces of Two Dice**

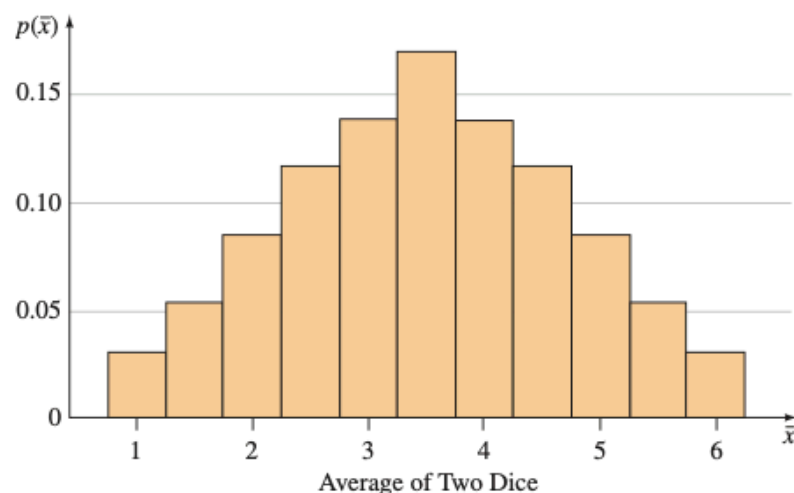
Second Die	First Die					
	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

■ **Table 7.5(b) Sampling Distribution of $\bar{x}$**

$\bar{x}$	$p(\bar{x})$	$\bar{x}$	$p(\bar{x})$
$2/2 = 1$	$1/36$	$8/2 = 4$	$5/36$
$3/2 = 1.5$	$2/36$	$9/2 = 4.5$	$4/36$
$4/2 = 2$	$3/36$	$10/2 = 5$	$3/36$
$5/2 = 2.5$	$4/36$	$11/2 = 5.5$	$2/36$
$6/2 = 3$	$5/36$	$12/6 = 6$	$1/36$
$7/2 = 3.5$	$6/36$		

Figure 7.4

Sampling distribution of $\bar{x}$
for $n = 2$ dice



Using a similar procedure, we generated the sampling distributions of $\bar{x}$ when $n = 3$ and $n = 4$. For $n = 3$, the sampling distribution in Figure 7.5 clearly shows the mound shape of the normal probability distribution, still centered at $\mu = 3.5$. Notice also that the spread of the distribution is slowly *decreasing* as the sample size n *increases*. Figure 7.6 dramatically

Figure 7.5

Sampling distribution of $\bar{x}$
for $n = 3$ dice

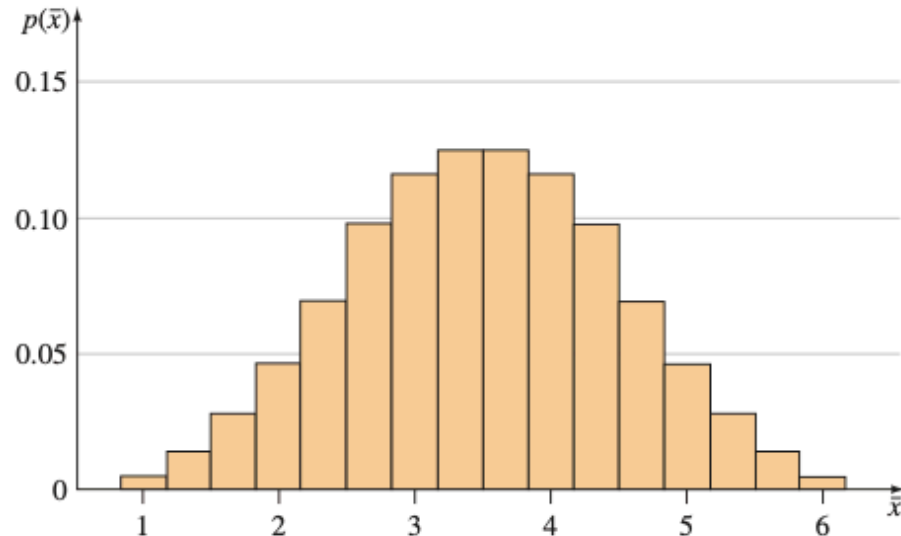
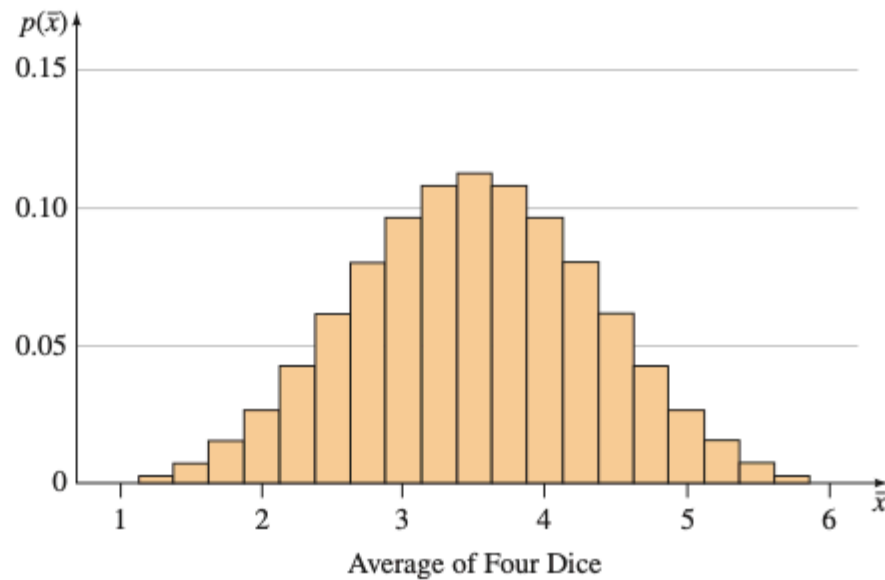


Figure 7.6

Sampling distribution of $\bar{x}$
for $n = 4$ dice



SAMPLING DISTRIBUTIONS

Sampling distributions for statistics can be

- ✓ Derived using mathematical theorems
- ✓ Approximated with simulation techniques (beyond this course)
- ✓ Approximated by the Central Limit Theorem.

Sampling with replacement

CENTRAL LIMIT THEOREM (SAMPLE MEAN)

1. Consider a sample of size n following a distribution $\mathcal{G} := \mathcal{G}(\text{mean} = \mu, \text{variance} = \sigma^2)$.
2. Suppose $X_1, \dots, X_n \sim^{iid} \mathcal{G}$ are identically and independently distributed (iid) with mean μ and variance σ^2 .

3. Define sample mean $\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n}$. Then, $E(\bar{X}_n) = \mu$,
 $\text{var}(\bar{X}_n) = \sigma^2/n$.

- The CLT says: when n is large ($n \geq 30$):

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \text{ as } n \rightarrow \infty$$

白話文重述中央極限定理

- Professional: The normalized sample mean converges to the standard normal random variable in distribution.
- Plain language: The normalized sample mean has a distribution approximately as a standard normal distribution.
- 專業：標準化後的樣本平均數，**分布收斂**到標準常態分配。
- 白話文：標準化後的樣本平均數，其分布很接近標準常態分配。

IN-CLASS EXERCISE

- Describe the Central Limit Theorem.
 - (1) Use math notations
 - (2) Use plain language

HOW LARGE IS LARGE?

If the sample is **normal**, then the sampling distribution of $\bar{x}$ will also be normal, no matter what the sample size.

When the sample population is approximately **symmetric**, the distribution becomes approximately normal for relatively small values of n .

When the sample population is **skewed**, the sample size must be **at least 30** before the sampling distribution of $\bar{x}$ becomes approximately normal.

The Sampling Distribution of the Sample Mean, $\bar{x}$

- If a random sample of n measurements is selected from a population with mean μ and standard deviation σ , the sampling distribution of the sample mean $\bar{x}$ will have mean μ and standard deviation

$$\frac{\sigma}{\sqrt{n}}$$

also called as the standard error.

Case 1

If the population has a *normal* distribution, the sampling distribution of $\bar{x}$ will be *exactly* normally distributed, *regardless of the sample size, n* .

Case 2

If the population distribution is *nonnormal*, the sampling distribution of $\bar{x}$ will be *approximately* normally distributed for large samples (by the Central Limit Theorem). Conservatively, we require $n \geq 30$.

THE SAMPLING DISTRIBUTION OF $\bar{X}_n$

- If a random sample of n measurements is selected from a distribution $\mathcal{G}$ with mean μ and standard deviation σ .
- Then, $\bar{X}_n$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

- If $\mathcal{F}$ is a normal distribution, we have

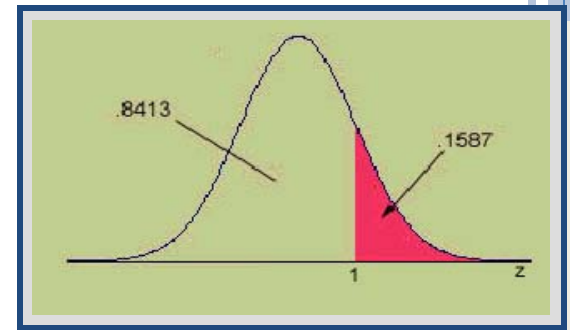
$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

常態分布的性質: ch10

- If $\mathcal{F}$ is not a normal distribution, the **CLT** says

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \text{ as } n \rightarrow \infty.$$

EXAMPLE: FINDING PROBABILITIES FOR THE SAMPLE MEAN



A random sample of size $n = 16$ from a normal distribution with $\mu = 10$ and $\sigma = 8$.

$$\begin{aligned} P(\bar{x} > 12) &= P\left(z > \frac{12 - 10}{8/\sqrt{16}}\right) \\ &= P(z > 1) = 1 - .8413 = .1587 \end{aligned}$$

EXAMPLE

A soda filling machine is supposed to fill cans of soda with 12 fluid ounces. Suppose that the fills are actually normally distributed with a mean of 12.1 oz and a standard deviation of .2 oz. What is the probability that the average fill for a 6-pack of soda is less than 12 oz?



Reiterate the problem

Population: $X \sim N(12.1, 0.2^2)$

Define $\bar{X}_6$ as the average fill for a 6-pack of soda.

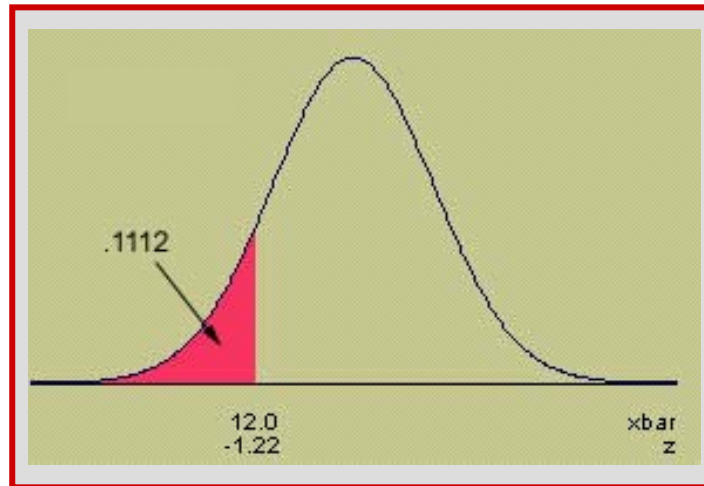
Find $P(\bar{X}_6 < 12)$

Key: $\frac{\bar{X}_6 - 12.1}{0.2/\sqrt{6}} \sim N(0,1)$

EXAMPLE



$$\begin{aligned} P(\bar{X}_6 < 12) &= P\left(\frac{\bar{X}_6 - 12.1}{0.2/\sqrt{6}} < \frac{12 - 12.1}{0.2/\sqrt{6}}\right) \\ &= P(Z < -1.22) = 0.1112 \end{aligned}$$



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7.4 ASSESSING NORMALITY

38

1. Histogram. Difficult to decide

2. Boxplot

- Can be used to check symmetry
- but not the tail behavior

3. Normal probability plot 

- Also known as Quantile-Quantile plot (QQ plot)
- X and y axis could be swapped from different computing programs.

HOW TO DRAW A NORMAL PROBABILITY PLOT

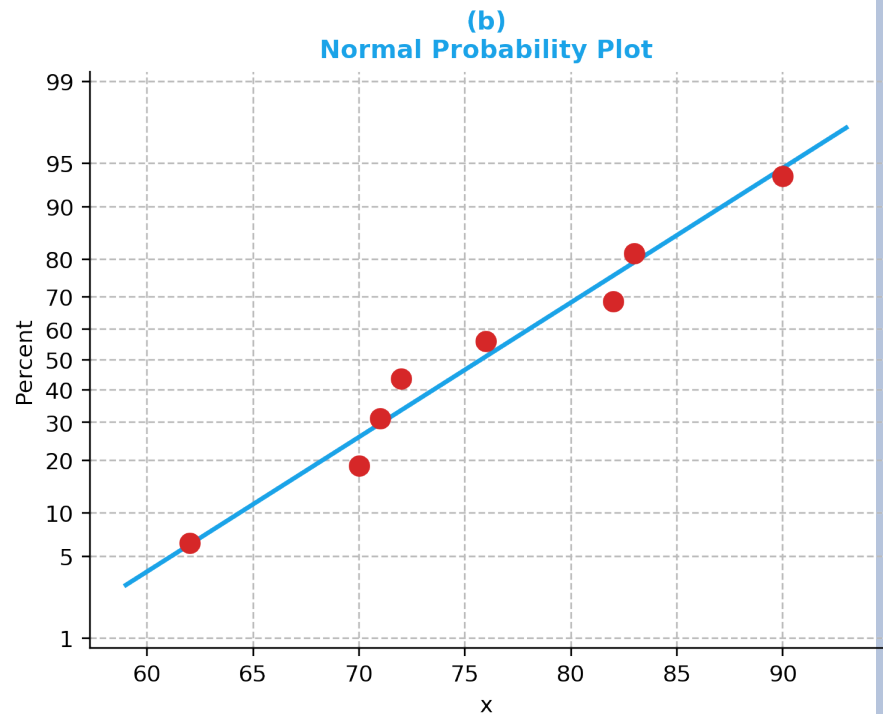
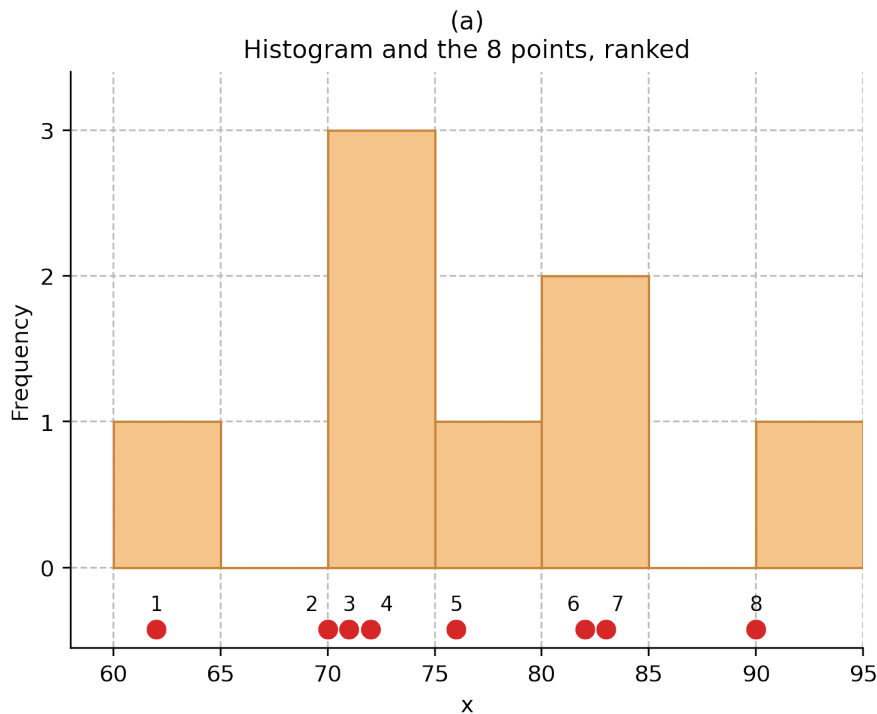


Figure made with npp-demo.py (03-KEYNOTE-particify-IC); data = [62, 70, 71, 72, 76, 82, 83, 90]

STEP BY STEP: THE 8 POINTS

rank i	x (ranked)	$p = (i - 0.5)/n$	$z = \Phi^{-1}(p)$	percent
1	62	0.0625	-1.53	6.2%
2	70	0.1875	-0.89	18.8%
3	71	0.3125	-0.49	31.2%
4	72	0.4375	-0.16	43.8%
5	76	0.5625	0.16	56.2%
6	82	0.6875	0.49	68.8%
7	83	0.8125	0.89	81.2%
8	90	0.9375	1.53	93.8%

1) Rank the data. 2) $p = (i - 0.5)/n$, then z from the z table. 3) Plot x against z . 4) Label the z axis in percent.

EXAMPLE 7.6

The histogram and normal probability plot in Figure 7.10 were constructed based upon a sample of $n = 50$ observations from a normal population with mean $\mu = 10$ and standard deviation $\sigma = 2$. Comment on the shape of the histogram and whether the normal probability plots can reasonably be described as a straight line.

Figure 7.10

Histogram and normal probability plot for data from a normal distribution

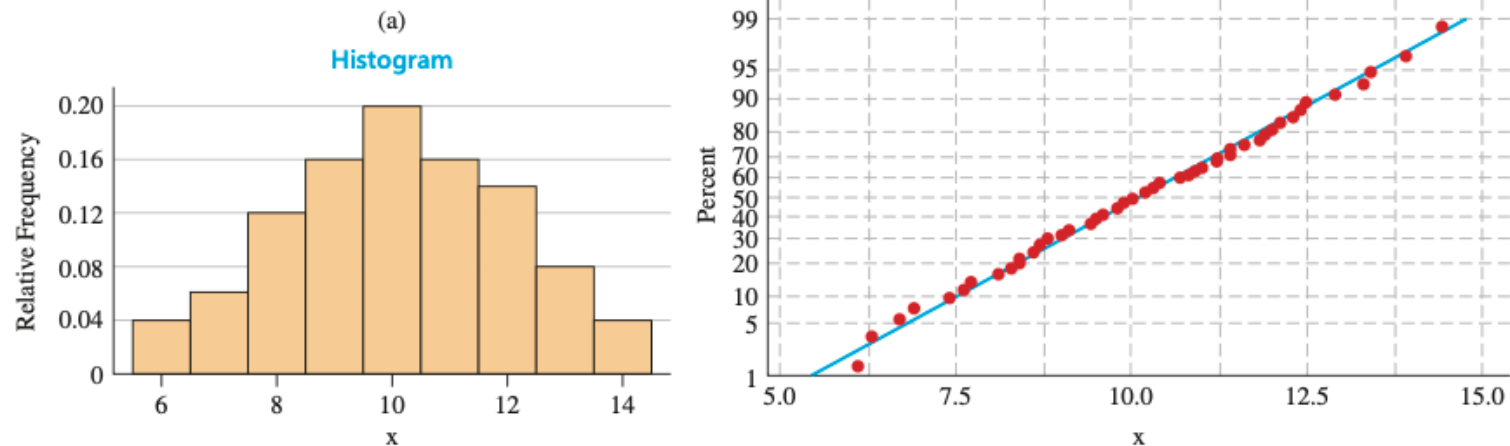
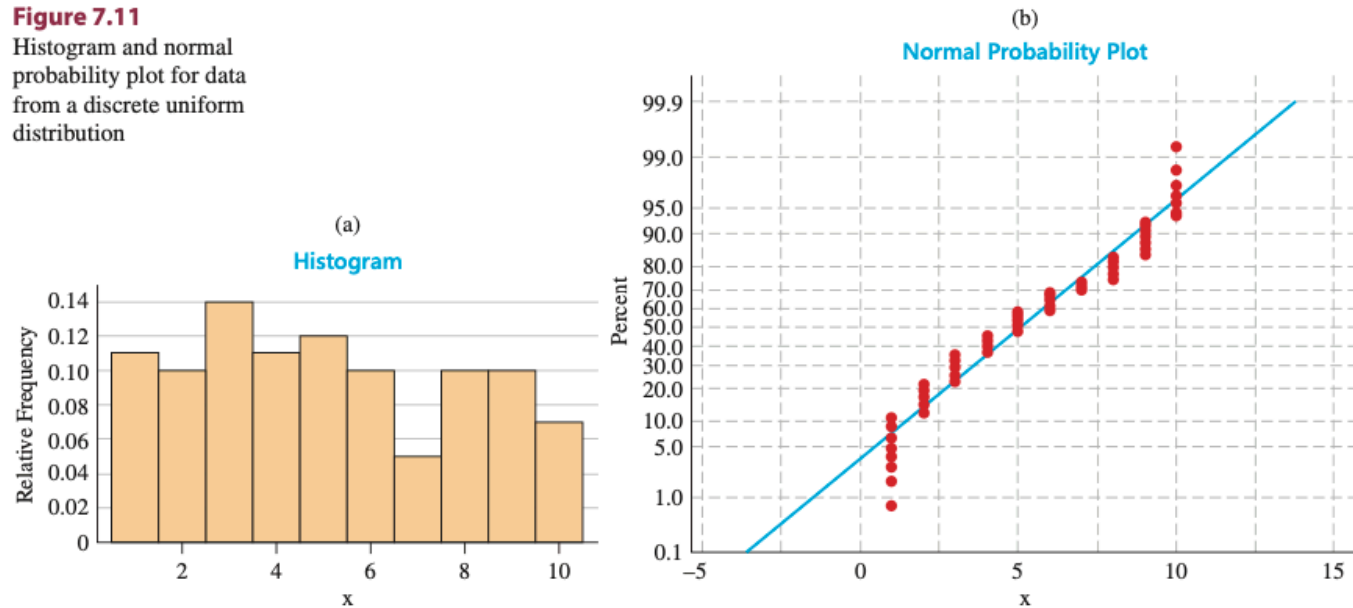


Figure 7.11

Histogram and normal probability plot for data from a discrete uniform distribution



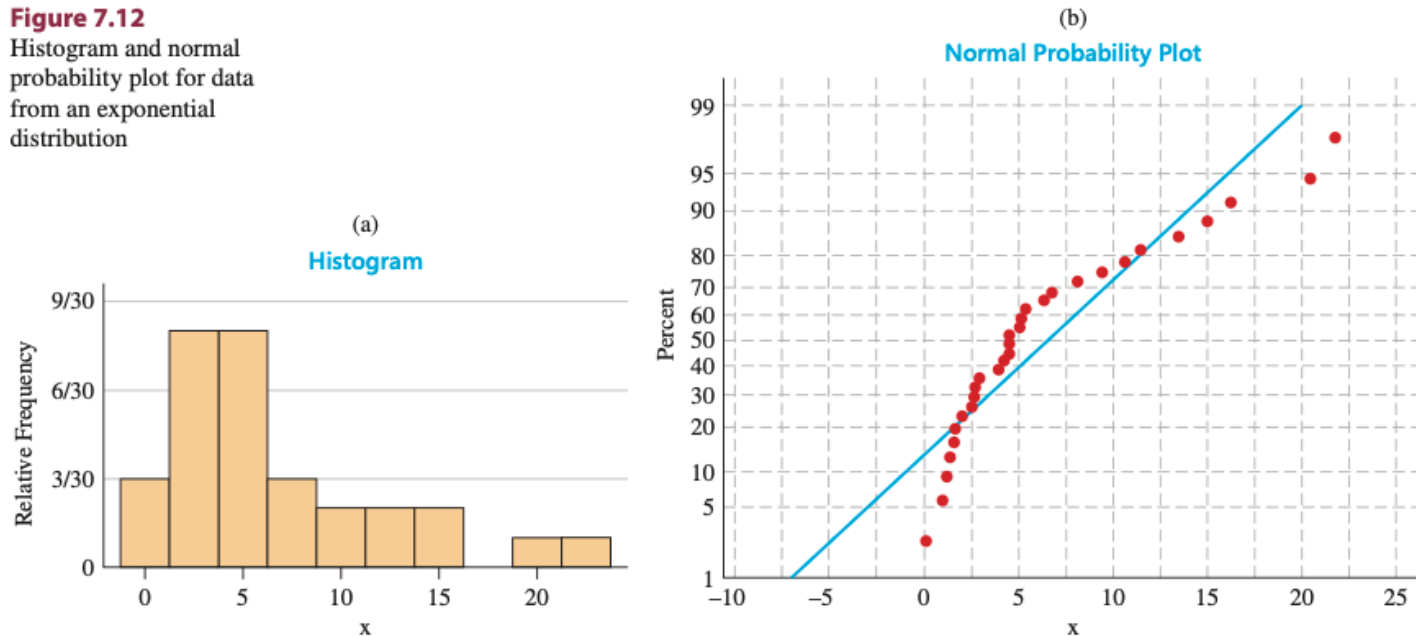
Solution The histogram is far from mound-shaped, and is relatively flat, characteristic of a discrete uniform distribution, and hence not normal. The normal probability plot shows a downturn in the lower area of the plot and an upturn in the upper area of the plot. This reflects the fact that the tails do not taper off like the normal curve, but rather, both tails are cut off, the lower at 1 and the upper at 10. This is *not* characteristic of the normal distribution.

EXAMPLE 7.8

Suppose a sample of $n = 30$ is selected from an exponential distribution with intensity $\lambda = 5$ where $\lambda = 1/\mu$. Compare the resulting histogram and normal probability plot in Figure 7.12 with those for a normal sample.

Figure 7.12

Histogram and normal probability plot for data from an exponential distribution

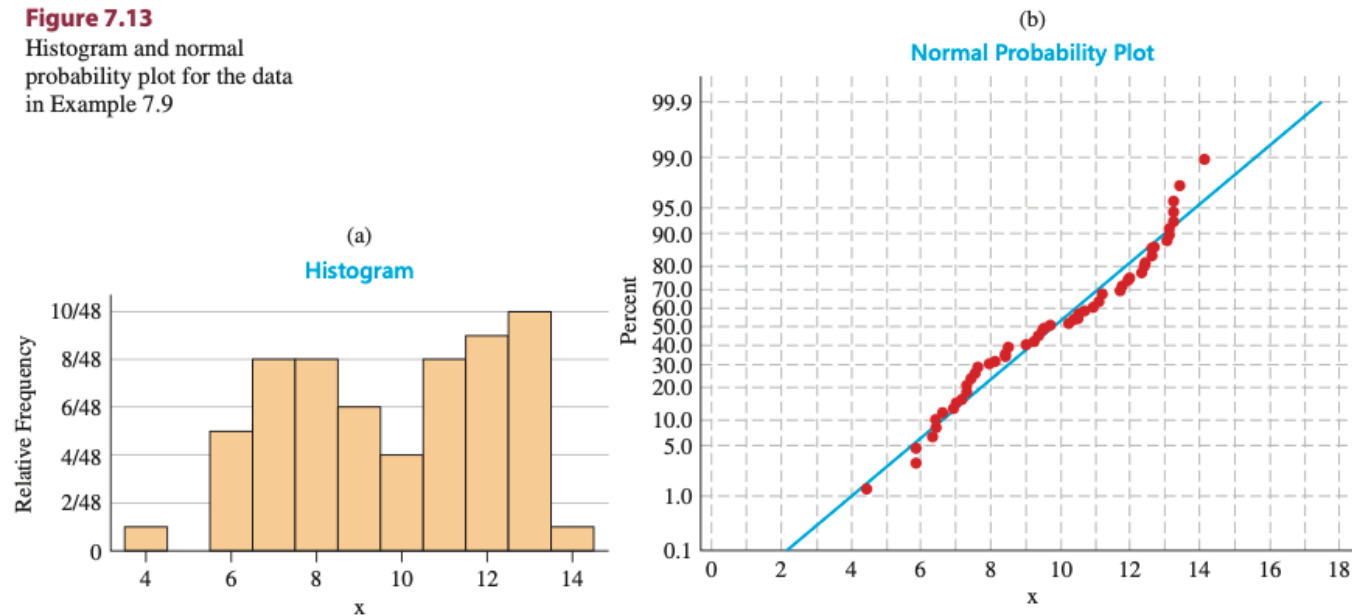


Solution The histogram is not mound-shaped, but rather is skewed to the right, displaying the shape of an underlying exponential distribution. Notice that the left tail of the normal probability plot dips downward at zero on the x scale reflecting the fact that x cannot be less than 0. Both graphs indicate that the underlying distribution is *not* normal.

EXAMPLE 7.9

The data are $n = 48$ sea-level pressures measured monthly for 4 years. Discuss the nonnormal aspects of the graphs in Figure 7.13.

Figure 7.13
Histogram and normal probability plot for the data in Example 7.9



Solution The data are not normal based upon the histogram in Figure 7.13(a). The probability plot in Figure 7.13(b) has the appearance of a wavy line first below the centerline, then above, again below ending above the centerline, indicating a periodic pattern in the data.

The left side of the slide features a series of vertical stripes in various shades of blue. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle contains the number 46.

7.5 THE SAMPLING DISTRIBUTION OF THE SAMPLE PROPORTION

46

Sampling with replacement

CENTRAL LIMIT THEOREM (SAMPLE MEAN)

1. Consider a sample of size n following a distribution $\mathcal{G} := \mathcal{G}(\text{mean} = \mu, \text{variance} = \sigma^2)$.
Bernoulli(p), $\mu = p, \sigma^2 = p(1 - p) = pq$
2. Suppose $X_1, \dots, X_n \sim^{iid} \mathcal{G}$ are identically and independently distributed (iid) with mean $\mu = p$ and variance $\sigma^2 = pq$.
3. Define sample mean $\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n} = \hat{p}$. Then, $E(\bar{X}_n) = \mu = p$,
 $\text{var}(\bar{X}_n) = \sigma^2/n = pq/n$.
- CLT: when n is large ($n \geq 30$):

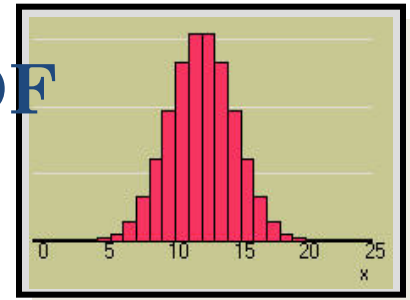
$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\hat{p} - p}{\sqrt{pq/n}} \xrightarrow{d} N(0,1) \text{ as } n \rightarrow \infty$$

SPECIAL CASE OF THE CLT

- Let $\mathcal{G}$ as *Bernoulli*(p).
 - *Bernoulli*(p) has mean p and variance pq
- The sample mean $\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n}$
 - Also called the sample proportion $\hat{p}$
 - $\hat{p}$ has the mean p and the variance $\frac{pq}{n}$.
 - $\hat{p}$ has the standard deviation $\sqrt{\frac{pq}{n}}$

The standard deviation of p -hat is sometimes called the STANDARD ERROR (SE) of p -hat.

THE SAMPLING DISTRIBUTION OF THE SAMPLE PROPORTION



- Assume $X_i \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$
- If n is large, and p is not too close to zero or one, By CLT,

$$\frac{\hat{p} - p}{\sqrt{pq/n}} \xrightarrow{d} N(0,1)$$

課本用 **Binomial(n, p)** 而非 **bernoulli(p)**

優點：不用特別介紹 **Bernoulli random variable**

缺點：看不出來是 **CLT** 的特例

- Suppose $X \sim \text{Binomial}(n, p)$.

$$\iff X = \sum_{i=1}^n X_i, \quad X_i \sim \text{Bernoulli}(p), i = 1, \dots, n$$

- The sample proportion is exactly the same mean:

$$\hat{p} = \frac{X}{n} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n.$$

- Thus, we have

$$\frac{\hat{p} - p}{\sqrt{pq/n}} \longrightarrow^d N(0, 1).$$

EXAMPLE #7.5.9

Random samples of size $n = 75$ were selected from a Binomial population with $p = 0.4$. Approximate $P(\hat{p} \leq 0.43)$ using CLT.

Sol. 

Model: $X_i \sim \text{Bernulli}(p)$, $i = 1, \dots, n$

$$\begin{aligned} P(\hat{p} \leq 0.43) &= P\left(\frac{\hat{p} - 0.4}{\sqrt{0.4 * 0.6/75}} \leq \frac{0.43 - 0.4}{\sqrt{0.4 * 0.6/75}}\right) \\ &\approx P\left(Z \leq \frac{0.43 - 0.4}{\sqrt{0.4 * 0.6/75}}\right) \\ &= P(Z \leq 0.53) = 0.70 \end{aligned}$$

EXAMPLE #7.5.27

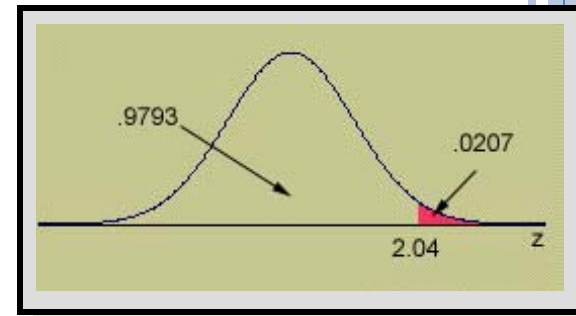
A fair coin is tossed $n = 80$ times. Let $\hat{p}$ be the sample proportion of heads. Find $P(0.44 < \hat{p} < 0.61)$.

Sol. 

Model: $X_i \sim \text{Bernoulli}(0.5)$, $i = 1, \dots, 80$.

$$\begin{aligned} P(0.44 < \hat{p} < 0.61) &= P\left(\frac{0.44 - 0.5}{\sqrt{0.5 \times 0.5/80}} < \frac{\hat{p} - 0.5}{\sqrt{0.5 \times 0.5/80}} < \frac{0.61 - 0.5}{\sqrt{0.5 \times 0.5/80}}\right) \\ &\approx P\left(\frac{0.44 - 0.5}{\sqrt{0.5 \times 0.5/80}} < Z < \frac{0.61 - 0.5}{\sqrt{0.5 \times 0.5/80}}\right) \\ &= P(-1.07 < Z < 1.96) = 0.8339. \end{aligned}$$

FINDING PROBABILITIES FOR THE SAMPLE PROPORTION



A random sample of size $n = 100$ from a binomial population with $p = .4$. Find the probability that sample proportion is at least 0.5.

Find $P(\hat{p} > 0.5)$.

Sol. 🧐

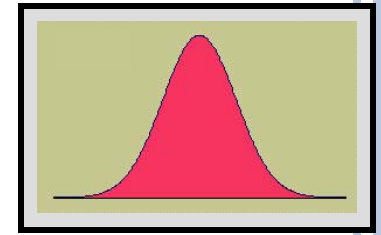
Model: $X_i \sim \text{Bernoulli}(p=0.4)$ for $i = 1, \dots, 100$

$$\begin{aligned} P(\hat{p} > 0.5) &= P\left(\frac{\hat{p} - 0.4}{\sqrt{0.4 \times 0.6/100}} > \frac{0.5 - 0.4}{\sqrt{0.4 \times 0.6/100}}\right) \\ &\approx P(Z > 2.04) \\ &= 1 - .9793 = 0.0207 \end{aligned}$$

The slide features a dark blue background with a series of vertical stripes in varying shades of blue on the left side. Several blue circles of different sizes are arranged vertically along these stripes. The main title is written in a white, serif, all-caps font.

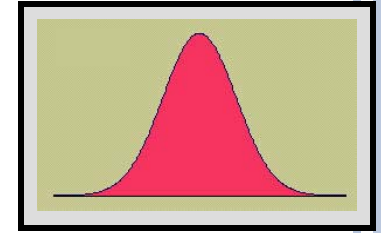
7.6 A SAMPLING APPLICATION: STATISTICAL PROCESS CONTROL

STATISTICAL PROCESS CONTROL



- ☐ The cause of a change in the variable is said to be **assignable** if it can be found and corrected.
- ☐ Other variation that is not controlled is regarded as **random variation**.
- ☐ If the variation in a process variable is solely random, the process is said to be **in control**.
- ☐ If out of control, we must reduce the variation and get the measurements of the process variable within specified limits.

THE $\bar{x}$ CHART FOR PROCESS MEANS

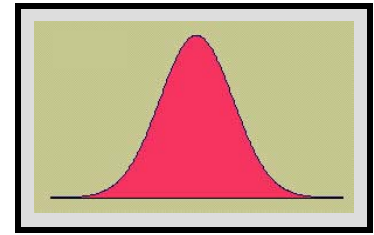


- ✓ At various times during production, we take a sample of size n and calculate the sample mean $\bar{x}$.
- ✓ According to the CLT, the sampling distribution of $\bar{x}$ should be approximately normal; almost all of the values of $\bar{x}$ should fall into the interval

$$\mu \pm 3 \frac{\sigma}{\sqrt{n}}$$

- ✓ If a value of $\bar{x}$ falls outside of this interval, the process may be out of control.

THE $\bar{\bar{x}}$ CHART

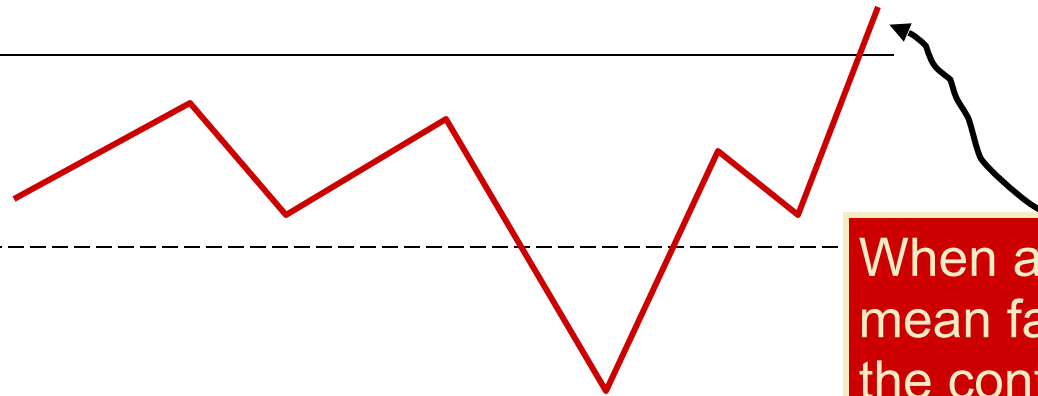
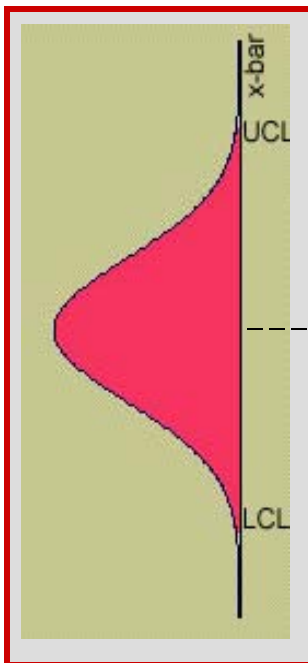


- ✓ To create a control chart, collect data on k samples of size n . Use the sample data to estimate μ and σ .
- ✓ The mean μ is estimated with $\bar{\bar{x}}$, the grand average of all the sample statistics calculated for the nk measurements on the process variable.
- ✓ The standard deviation σ is estimated by s , the standard deviation of the nk measurements.
- ✓ Create the control chart, using a **centerline** and **control limits**.

THE $\bar{x}$ CHART

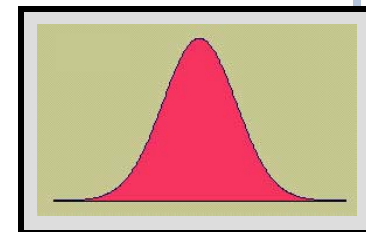
Centerline: $\bar{\bar{x}}$

$$\text{LCL: } \bar{\bar{x}} - 3 \frac{s}{\sqrt{n}} \quad \text{UCL: } \bar{\bar{x}} + 3 \frac{s}{\sqrt{n}}$$



When a sample mean falls outside the control limits, the process may be out of control.

THE P CHART FOR PROPORTION DEFECTIVE

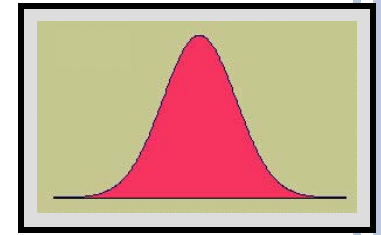


✓ At various times during production, we take a sample of size n and calculate the proportion of defective items, $\hat{p} = x/n$

✓ According to the CLT, the sampling distribution of $\hat{p}$ should be approximately normal; almost all of the values of $\hat{p}$ should fall into the interval

$$p \pm 3\sqrt{\frac{pq}{n}}$$

✓ If a value of $\hat{p}$ falls outside of this interval, the process may be out of control.



THE P CHART

✓To create a control chart, collect data on k samples of size n . Use the sample data to estimate p .

✓The population proportion defective p is estimated with

$$\bar{p} = \frac{\sum \hat{p}_i}{k}$$

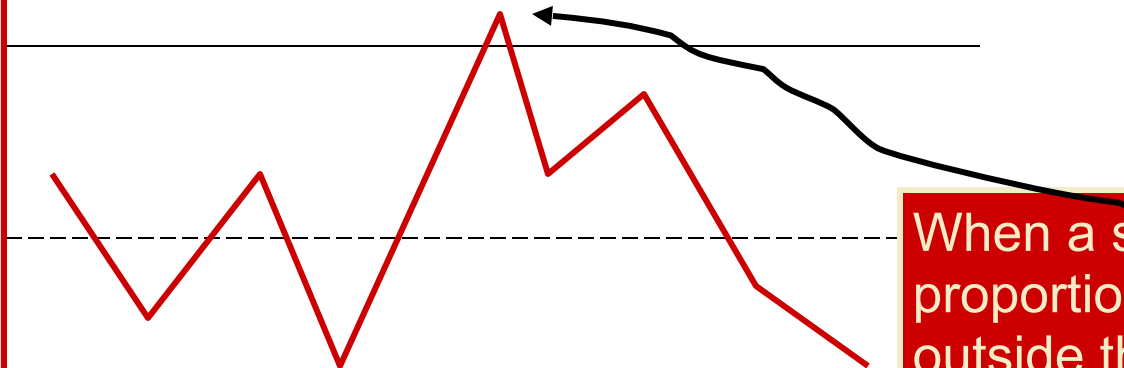
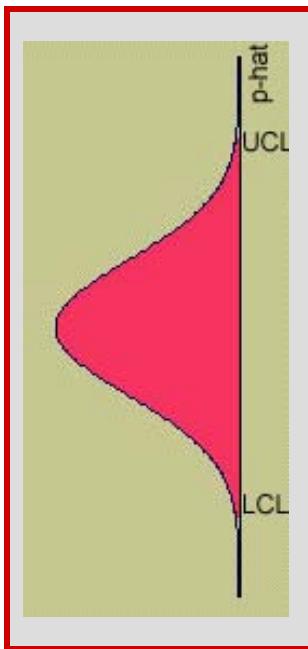
✓the grand average of all the sample proportions calculated for the k samples.

✓Create the control chart, using a **centerline** and **control limits**.

THE P CHART

Centerline: $\bar{p}$

$$\text{LCL} : \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \quad \text{UCL} : \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$



When a sample proportion falls outside the control limits, the process may be out of control.

The left side of the slide features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several blue circles of different sizes. One large circle is positioned near the top left, and several smaller circles are scattered below it, some overlapping the stripes.

CHAPTER REVIEW

62

KEY CONCEPTS

I. Sampling Plans and Experimental Designs

1. Simple random sampling
 - a. Each possible sample is equally likely to occur.
 - b. Use a computer or a table of random numbers.
 - c. Problems are nonresponse, undercoverage, and wording bias.
2. Other sampling plans involving randomization
 - a. Stratified random sampling
 - b. Cluster sampling
 - c. Systematic 1-in- k sampling

KEY CONCEPTS

3. Nonrandom sampling

- a. Convenience sampling
- b. Judgment sampling
- c. Quota sampling

II. Statistics and Sampling Distributions

1. Sampling distributions describe the possible values of a statistic and how often they occur in repeated sampling.
2. Sampling distributions can be derived mathematically, approximated empirically, or found using statistical theorems.
3. The **Central Limit Theorem** states that sums and averages of measurements from a nonnormal population with finite mean μ and standard deviation σ have approximately normal distributions for large samples of size n .

KEY CONCEPTS

III. Sampling Distribution of the Sample Mean

1. When samples of size n are drawn from a normal population with mean μ and variance σ^2 , the sample mean $\bar{x}$ has a normal distribution with mean μ and variance σ^2/n .
2. When samples of size n are drawn from a nonnormal population with mean μ and variance σ^2 , the Central Limit Theorem ensures that the sample mean $\bar{x}$ will have an approximately normal distribution with mean μ and variance σ^2/n when n is large ($n \geq 30$).
3. Probabilities involving the sample mean $\bar{x}$ can be calculated by standardizing the value of $\bar{x}$ using

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

KEY CONCEPTS

IV. Sampling Distribution of the Sample Proportion

1. When samples of size n are drawn from a $\hat{p}$ binomial population with parameter p , the sample proportion will have an approximately normal distribution with mean p and variance pq/n as long as $np > 5$ and $nq > 5$.
2. Probabilities involving the sample proportion can be calculated by standardizing the value using $\hat{p}$

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

KEY CONCEPTS

V. Statistical Process Control

1. To monitor a quantitative process, use an $\bar{x}$ chart. Select k samples of size n and calculate the overall mean $\bar{\bar{x}}$ and the standard deviation s of all nk measurements. Create upper and lower control limits as

$$\text{LCL: } \bar{\bar{x}} - 3 \frac{s}{\sqrt{n}} \quad \text{UCL: } \bar{\bar{x}} + 3 \frac{s}{\sqrt{n}}$$

If a sample mean exceeds these limits, the process is out of control.

2. To monitor a binomial process, use a p chart. Select k samples of size n and calculate the average of the sample proportions as

$$\bar{p} = \frac{\sum \hat{p}_i}{k}$$

Create upper and lower control limits as

$$\text{LCL: } \bar{p} - 3 \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \quad \text{UCL: } \bar{p} + 3 \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

If a sample proportion exceeds these limits, the process is out of control.